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	The overall objective of this NASA Specialized Center of Research (NSCOR) is to model the relative risk of gastrointestinal (GI) tumorigenesis for high priority space radiation beams and compare with gamma radiation where human epidemiologic data are available. It will rely primarily on mouse models of gastrointestinal (GI) cancer, and human colonic epithelial cell (HCEC) culture approaches that have been optimized for low dose exposures. This will involve quantitative and qualitative analyses of tumors as well as normal GI epithelium after radiation in mouse models. Results will be further extrapolated to identify radiation-induced molecular signatures of carcinogenesis in both mouse models and state-of-the-art 2D and 3D cultured human colonocytes. While limited studies at select loci have shown high energy (HZE)-induced mutations including splicing errors, frameshifts, indels and complex changes, we will carry out genome-wide analysis using the power of modern next-gen sequencing (NGS) to assess the spectrum of mutations in radiation-induced tumors, and also persistent genome-wide epigenomic changes in normal cells after ionizing radiation (IR). These NGS and epigenomics findings will be integrated with transcriptomics (RNAseq and/or microarray) and metabolomics data as well as with the select events at the molecular level to develop a systems biology model for GI tumorigenesis. By assessing both quantitative in vivo tumorigenesis data and molecular events in mouse tissues and human colonocytes, space radiation-induced tumorigenesis will be compared to human epidemiologic data for low and high LET (linear energy transfer) radiation exposures, and approaches will be developed to model human risks for GI cancer by space radiation. In addition to obtaining quantitative tumorigenesis data we also seek to test potential GI-cancer prevention agents (such as metformin and CDDO) to achieve desired risk mitigation against space radiation-induced tumorigenesis. We continue to employ two models of GI cancers: the APC1638N/+ model, which was generated by targeted modification of the Apc gene at a position corresponding to aa1638. These mice develop very few adenomas in the small as well as in the large intestine but post-radiation tumor numbers show marked increases. Radiation also induced substantial gastric tumorigenesis in this model. The second model is the CDX2P Apc flox/+ that has one APC allele inactivated almost exclusively in the colon leading to adenomas and subsequent low rate of adenocarcinomas in the colon that is increased post-irradiation. Data being collected from these two relevant mouse models will be used to calculate risk estimates for NASA. Since low-LET risk estimates are available for human populations, we will then be able to model the relative risk for space radiation. In particular, this ratio can then be extrapolated to human exposure to low-LET radiation to then make an estimate of risk for colorectal cancer (CRC) in humans exposed to space radiation compared to the standard risks based on atomic bomb survivors and other exposure populations.
Task Description:	In order to answer the critical question of risk for colonic tumorigenesis after acute and chronic space radiation exposures, in vitro human cell culture and animal models for CRC initiation and progression are needed. We are investigating the effects of high-LET particle and high-energy protons in colonic epithelial cells with respect to the nature of damage and consequent tumor induction using biological cancer-related endpoints and systems biology (genomics, proteomics, and metabolomics) approaches. In addition, we are comparing these results to the biological consequences of low dose rates and mixed fields of radiation which astronauts are likely to be continuously exposed. The studies of low- and high-LET radiation on colonocytes gain more relevance with increasing age when premalignant adenomas and other colonic lesions such as flat lesions and microscopic aberrant crypt foci become more prevalent. Pre-existing undetectable genetic or chromosomal aberrations in astronauts in addition to de novo mutations resulting from long-term irradiation may increase the chances of colonic stem cell malignant transformation. Plans for this proposal build on our past accomplishments with GI tumorigenesis models and are focused on studies at low doses (5 to 50 cGy) with an expanded range of NASA's high priority beams. This proposal is also focused on assessing the effects of gender, dose-rate, age, and genetic heterogeneity in mouse models. Molecular events triggered by space radiation will be investigated in-depth and the mechanistic basis for persistent aberrant cell signaling after HZE radiation will be dissected. Genome-wide NGS combined with other omics approaches will contribute to a systems biology understanding of the effects of space radiation in the intestines and stomach. Both quantitative murine tumor data and results for underlying molecular events in mouse and human GI cells will contribute to the modeling of human risks for GI cancer by space travel. The overall aims of this NSCOR are summarized below: Aim 1 (Project 1). Quantitatively assess GI tumorigenesis in mouse models of GI cancer and collect samples for qualitative analysis. Aim 2 (Project 2). Dissection of the signaling events and consequences in GI cells of the persistent effects of space radiation. Aim 3 (Project 3): Characterization of IR-induced neoplastic events in normal diploid human colonocytes. Aim 4 (Project 4). Development of systems biology (Project 4A) and mathematical modeling (Project 4B) approaches for GI cancer risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S, and the American Cancer Society estimates that there will be 147,950 new cases of colorectal cancer (CRC) with 53,200 persons predicted to die of the disease in 2020. Overall, the lifetime risk of developing colorectal cancer is: about 1 in 23 (4.4%) for men and 1 in 25 (4.1%) for women. Considering the high frequency of colorectal tumors in the American population, and an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. In addition to risk estimation, studies on the persistence of oxidative and inflammatory stress after HZE radiation and its role in driving gastrointestinal tumorigenesis may provide insight into mutagenic processes affecting genome integrity and carcinogenesis. The significance and deliverable of this project is to improve the estimates of GI cancer risk in astronauts, and to identify and test plausible targets for the development of mitigation strategies.

Task Progress:

In the case of APC1638N/+ mice, we are using this model to obtain quantitative and qualitative tumorigenesis data at space relevant doses and beams spanning a range of LET and energies as well as with mixed field beams. In 2019, we used the NASA Space Radiation Laboratory (NSRL) facility for two-beam runs in the summer (protons and 28Si-ion) and fall (28Si-ion) and we have exposed APC1638N/+, wild-type C57BL6/J, and Lgr5 (Lgr5-EGFP-IRES-creERT2) mice for GI-tumorigenesis, persistent stress signaling and risk mitigation studies. Our previous data in APC mutant mouse model demonstrated higher intestinal tumorigenesis after exposure to 12C, 28Si, and 56Fe radiation relative to gamma rays. We also observed important qualitative differences between heavy ions and gamma rays in both normal GI epithelial cells and tumors. Further additions to the quantitative data obtained until 2018 was done in 2019 using more mice, additional beam types (proton and mixed-field GCR), and doses (5 and 10 cGy) in male and female APC1638N/+ mice exposed to gamma-rays, proton (1000 MeV/n), 4He (250 MeV/n), 12C (290 MeV/n), 16O (325 MeV/n), 28Si (300 MeV/n), 56Fe (1000 MeV/n), and mixed-field GCR (60% proton, 20% 4He, 10% 16O and 10% 28Si –ions). Since APC1638N/+ mice are also a model of gastric tumorigenesis, we also compared gastric tumorigenesis after gamma-rays and 28Si-ion exposure, where we found a significant increase in gastric tumorigenesis after HZE exposure. Additionally, gastric tissues sections are being analyzed for histological signs of tumorigenesis to add more doses and beam types. Further, in consultation with NASA, we also carried out a pilot study to assess the potential protective effect of metformin (an anti-diabetic and anti-aging agent), which is also associated with decreased frequency of human CRC. While metformin has been reported to decrease aging phenotype and CRC risk in humans, the experimental approach was developed to test effectiveness of metformin in preventing/decreasing 28Si-induced GI-tumorigenesis in APC1638N/+ mice.
Task book accomplishments using APC1638N/+ and Lgr5 (Lgr5-EGFP-IRES-creERT2) mice:

1. Proton tumorigenesis study: In our earlier proton beam studies we have compared results of acute vs. fractionated exposure at high doses (1.2 Gy vs 0.4 Gy x3 fractions). Here, we exposed male and female APC1638N/+ to 50 cGy of proton radiation and tumors in the GI tract were counted after 150 days post-exposure. Further, proton tumorigenesis data was compared to the already acquired GI-tumorigenesis data after 50 cGy exposure of 4He, 12C, 16O, 28Si, and 56Fe –ions to understand radiation quality effects on GI-tumorigenesis.

2. Qualitative analysis of HZE-induced tumors: HZE-ion exposure led to higher carcinoma relative to gamma-rays. The highest % of carcinoma was observed after 56Fe radiation (25% were carcinoma) at 10 cGy dose. A similar trend of highest percentage pf carcinoma after 56Fe radiation was also observed after 50 cGy as well as after 2 Gy equitoxic doses of heavy-ion radiation.

3. Gastric tumorigenesis study: Male APC1638N/+ mice were whole-body exposed to sham-radiation, gamma-rays, and 28Si (300 MeV/n) in summer-2019 and mice were sacrificed150 days after irradiation, and gastric tumor-frequency was scored. A significant increase in gastric tumorigenesis after 10 cGy of 28Si (~3 fold higher) was observed compared to the same dose of gamma-rays.

4. Mixed-field GI tumorigenesis study in APC1638N/+ mice: Male and female APC1638N/+ mice were whole-body exposed to sham-radiation and mixed-field GCR type beam in fall of 2018 and mice were sacrificed 150 days after irradiation, and gastrointestinal tumor-frequency was scored. We compared quantitative tumor data obtained from the equivalent doses of single HZE beam to the mixed-field GCR-induced tumorigenesis that revealed a higher relative contributions of the dose received from HZE.

5. Persistent effects study: Lgr5+ mice (6 to 8 weeks) were exposed to 50 cGy of 28Si-ions and intestinal stem cells were isolated through flow-sorting and markers of persistent stress, senescence, and SASP are being analyzed. Additionally, these samples are also being used for next-gen transcriptomics (RNAseq) and epigenomic studies.

6. Metformin risk mitigation study: Tested risk mitigation efficacy of metformin (0.1% in diet). The male APC1638/+ mice were put on metformin diet at 4 weeks of age and continued on metformin to the end of the study. Mice were exposed to 28Si-ion (10 cGy) at 8 week of age and GI-tumorigenesis was estimated at 150 days post-exposure. Metformin administration led to a statistically significant reduction in HZE-induced GI-tumorigenesis.

In addition to our primary focus on GI-tumorigenesis using APC-based mouse models (APC1638N/+ and CDX2P Apc flox/+), in consultation with NASA, we also added a lung cancer mouse model (LA1) to study space radiation-induced lung tumorigenesis and mitigation study using CDDO.

Task book accomplishments using CDX2P Apc flox/+ and LA1 mice:

1. Simulated solar particle events (sSPE) radiation induces prolonged DNA damage, and senescence-associated inflammation in colon tissues.

2. sSPE increases colon cancer initiation and progression in CPC;APC mice. Increased p53 mutations in colon cancer after sSPE exposure was also observed.

3. SPE-irradiation results in a ~3-fold increased incidence of colon carcinoma in the CPC;APC mouse model while administration of CDDO diet during irradiation exhibit fewer polyps in comparison to mice on control diet.

4. CDDO provided 3 days prior to radiation exposure protected mice from sSPE induced colon cancer initiation and progression. The CPC;APC mice on CDDO diet during solar particle irradiation simulations exhibit a decreased incidence in invasive carcinoma in comparison to mice on control diet.

5. 5cGy (300 MeV/n) of a single dose exposure to Silicon does not increase cancer initiation or overall survival but 10 cGy results in a >2-fold cancer initiation and an overall decrease in lifespan.

6. LA1 mice on CDDO diet during fractionated 56Fe- irradiation exhibit a decreased incidence in invasive carcinoma in comparison to mice on control diet.

7. LA1 mice on CDDO diet during solar particle irradiation simulations exhibit a decreased incidence in invasive carcinoma in comparison to mice on control diet.

8. LA1 mice fed CDDO without irradiation show reduced tumor progression.

9. Mixed beam studies were also carried out in CDX2P Apc flox/+ mice using both acute and protracted GCR sim beans.

Task book accomplishments on genetic and epigenetic analysis of space radiation tumor and normal tissues: Using next-gen sequencing approaches, we defined the features of the tumor genomes arising in our murine models after

gamma, proton, and HZE exposure. During 2019, efforts have focused on two primary areas described in detail below:

1. DNA sequence analysis of mouse colon tumor samples arising after radiation exposure: Using our well established next-generation sequencing pipeline, we first used a targeted approach to sequence the Apc and Trp53 regions. From targeted sequencing, there is evidence that the loss of the Apc region is the primary mechanism of tumor initiation. However, inactivating mutations in Apc without LOH have been identified in some tumors from Si irradiated mice. These results have been validated with the sensitive digital droplet PCR assay. At a whole-genome scale, we will have the potential to detect differences in the mutation signature between these groups including the presence of structural rearrangements as well as small mutations. We have now obtained whole-genome sequencing (WGS) on some samples and are in the process of obtaining WGS on a collection of tumors from control mice and those exposed to gamma or 28Si.

2. Development of a targeted bisulfite-seq platform for the epigenetic analysis of irradiated mouse tissue: It has now been firmly established that HZE radiation leads to a persistent disturbance in the architecture of the gut epithelium. To investigate the mechanistic basis of this apparently epigenetic phenomenon, we are utilizing the Lgr5-GFP model established in the Fornace lab in addition to the Apc cancer-prone models. However, there are limited commercial tools for large-scale analysis of DNA methylation in the mouse genome. While whole-genome bisulfite sequence is possible, it is prohibitively resource-intensive. Our solution is the implementation of a large-scale targeted bisulfite sequencing platform capturing 69.9 Mb with 128,614 oligonucleotides. This provides a reasonable sampling of the mouse genome. After validating this system with high molecular weight DNA, we have initiated the analysis of DNA samples extracted from fixed tissues harvested from experimental and control mice. Results are excellent with clear delineation of CpG methylation status at the nucleotide level. We are now processing 24 samples and beginning the search for differentially methylated regions that may be related to radiation exposure.

Task book accomplishments on risk modeling using mouse tumorigenesis data:

Astronauts embarking on a trip to Mars and back or on other lengthy space exploration missions will encounter a multi-component mixture of ionizing radiations. These radiations damage cells in multiple ways, which can be roughly classified into two categories:

1. Targeted effects (TE), involving the consequences of direct traversals of cells by ionizing tracks leading to DNA double-strand breaks and other lesions.

2. Non-targeted effects (NTE, also called “bystander” effects) caused by the release of signals from cells directly hit by the tracks and the effects of these signals on other cells.

The presence of both TE and NTE can produce complex dose response shapes for space radiations, particularly in the low dose region that is most relevant for space missions.

Mechanistically-motivated radiation carcinogenesis models serve as important tools to describe these dose responses and make estimates and predictions of cancer risks from space exploration. Here we used our model that includes both TE and NTE to analyze updated mouse tumorigenesis data from the current NSCOR. The updated data set now includes more detailed information on space-relevant doses of multiple radiation types. A more detailed modified model formalism that uses a weighted Negative Binomial distribution to better describe the data variability was developed and implemented on this data set. This new information and improved modeling approach allowed a better estimation of dose-response shapes in the space-relevant dose region. Specifically, NTE parameters and relative biological effectiveness (RBE) and radiation effects ratio (RER) values at such doses of heavy ions and protons were estimated with more information than previously possible. Ultimately, these improvements provided updated and enhanced estimation of gastrointestinal cancer risks in human astronauts.

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